

Energy-Flow Cosmology, Part 4: Regime Susceptibility and Cross-Scale Mapping

Morten Magnusson

Symbiose Research, Sandnes, Norway

ORCID: 0009-0002-4860-5095

Preprint — April 2026

DOI: 10.6084/m9.figshare.31970907

Abstract. Parts 1–3 of this series established EFC as a perturbation-sector modification of the growth sector with correct limiting behaviour and a persistent empirical signal. This fourth paper addresses the open problem of connecting the galactic and cosmological scales within a single framework. We propose a *regime susceptibility function* $T(S) = S_0(1 - S_0)/[S(1 - S)]$ that emerges directly from the $S(1 - S)$ structure already present in EFC — no new fields or parameters are introduced. $T(S)$ provides a parameter-free cross-scale mapping from the SPARC-calibrated coupling $\beta_0 \approx 0.16$ to a cosmological amplitude giving $w_0 \approx -1$, while the product $\beta(S) \cdot \rho_{\text{eff}}(S) = \text{const}$ ensures exact self-regulation and GR recovery at $S \rightarrow 0$. Dark matter and dark energy are reinterpreted as the convergent and divergent phases of a single entropic continuum; their observed magnitudes arise from regime-mismatch in the measurement chain. The amplification factor $\beta_{\text{cosm}}/\beta_{\text{gal}} \approx 6.25$ is a measurable, parameter-free prediction. The dynamical dark energy prediction $w(a) \neq -1$ is directly testable by Euclid (Kill Criterion 5). All results are labelled as candidate mechanisms requiring derivation from first principles.

Companion papers: Part 1 (Recovery & Limits), [figshare.31970886](#); Part 2 (Field Equations & Observables), [figshare.31970898](#); Part 3 (Validation & Falsification), [figshare.31970904](#).

Introduction

Parts 1–3 of the EFC whitepaper series established:

1. Recovery of Λ CDM/GR under three independent conditions (Part 1).
2. A complete observable mapping from entropy field $S(a)$ to $f\sigma_8$, S_8 , Σ , and $R(k, S)$ (Part 2).
3. A structured validation ledger and sharp kill criteria for Stage-IV surveys (Part 3).

One structural gap remained: EFC operates in the *perturbation sector only*. No consistent mapping to the background expansion $H(z)$ was established, and the coupling amplitude β was calibrated at galactic scales (SPARC) without a derivation of its cosmological counterpart.

This paper addresses both gaps through a single conceptual move: the recognition that $S(1 - S)$, already central to EFC’s effective energy density, generates a natural *regime susceptibility* that maps couplings across scales.

We do not claim to have derived $H(z)$ from first principles. We claim to have identified the structural element — $T(S)$ — that any such derivation must contain.

The Entropic Continuum

Dark matter and dark energy as phases, not substances

EFC treats the universe as a thermodynamically closed system in which a single current J^μ flows along entropy gradients. The divergence of this current defines two dynamical phases:

$$\nabla_\mu J^\mu \begin{cases} < 0 & \text{convergent flow} \rightarrow \text{excess gravity (DM regime)} \\ = 0 & \text{equilibrium} \rightarrow \text{matter-dominated epoch} \\ > 0 & \text{divergent flow} \rightarrow \text{accelerated expansion (DE regime)} \end{cases} \quad (1)$$

This is not a restatement of Λ CDM with relabelled components. It is a structural claim: the apparent need for two independent dark substances arises from applying a single-regime ontology (particle physics) to a multi-regime process (entropic flow).

The thermodynamic boundaries

The entropy field $S \in [0, 1]$ spans two limiting states:

Definition 1 (Singularity). $S \rightarrow 0$: maximal order, zero entropy gradient, no energy flow. The universe is in its most structured, dynamically inert state. GR is recovered exactly (Part 1, Theorem 2).

Definition 2 (Altular limit). $S \rightarrow 1$: maximal entropy, zero gradient, no flow. The universe approaches thermodynamic equilibrium. Expansion asymptotes to de Sitter; all coupling vanishes.

Between these boundaries, the universe evolves monotonically: $dS/dt > 0$ always. The ratio of divergent to convergent flow — empirically identified as Ω_Λ/Ω_m in Λ CDM — is not a constant but a snapshot of this evolution:

$$\frac{\Omega_\Lambda(z)}{\Omega_m(z)} = f\left(\frac{S(z)}{S_{\max}}\right), \quad (2)$$

where f is monotonically increasing. The observed value $\Omega_\Lambda/\Omega_m \approx 2.5$ at $z = 0$ is therefore a measurement of where the universe currently sits on the entropic continuum, not a fundamental cosmological constant.

Effective Energy Density and Pressure

$$\rho_{\text{eff}} = S(1 - S)$$

The effective energy density available for dynamical exchange is:

$$\rho_{\text{eff}}(S) = S(1 - S). \quad (3)$$

This form is uniquely motivated:

- $S \rightarrow 0$: $\rho_{\text{eff}} \rightarrow 0$. No energy available at the singularity.
- $S \rightarrow 1$: $\rho_{\text{eff}} \rightarrow 0$. No energy available at the Altular limit.
- $S = 0.5$: $\rho_{\text{eff}} = 0.25$, maximum. Peak dynamical activity at the midpoint.

Equation (3) has the form of a Bernoulli variance — the information-theoretic capacity of a binary channel. In statistical physics this is the susceptibility of a system near a phase transition. EFC inherits this structure from the thermodynamic continuum.

Effective pressure and the equation of state

The effective pressure is proportional to the divergence of the energy-flow current:

$$p_{\text{eff}} \propto -\nabla_{\mu} J^{\mu}. \quad (4)$$

Convergent flow ($\nabla_{\mu} J^{\mu} < 0$) gives positive effective pressure — structure resists compression. Divergent flow ($\nabla_{\mu} J^{\mu} > 0$) gives negative effective pressure — the system “releases” rather than “pushes.”

Accelerated expansion in EFC is therefore not driven by an energy injection. It is the *relaxation of a binding constraint* as entropy approaches its maximum.

The effective equation of state follows from $p_{\text{eff}}/\rho_{\text{eff}}$, where $\rho_{\text{eff}} = S(1 - S)$ cancels in the ratio:

$$w(a) \equiv \frac{p_{\text{eff}}}{\rho_{\text{eff}}} = -\beta(S) \cdot a, \quad (5)$$

where $\beta(S) = \beta_0 T(S)$ is the regime-scaled coupling amplitude defined in Section 4. The sigmoid sharpness n controls the *shape* of the entropy transition; it is independently constrained by the growth sector ($n \sim 2\text{--}4$, Parts 1–3) and does not enter the amplitude w_0 . This separation is essential: n and $\beta(S)$ are independent quantities with distinct observational constraints.

- Early universe ($a \ll 1$, $S \approx 0$): $\beta(S) \approx \beta_0$, $w \approx 0$ — matter-like.
- Today ($a = 1$): $w_0 = -\beta_{\text{cosm}}$, determined by S_{cosm} (Section 5).
- Altular limit ($S \rightarrow 1$): $\beta(S) \cdot S(1 - S) = \text{const}$ (Proposition 1), $w \rightarrow -1$ asymptotically — de Sitter.

Remark 1 (Dynamical dark energy — no parameter recycling). Equation (5) predicts $w(z) \neq -1$. The growth-sector parameter n is not reused here. The amplitude $w_0 = -\beta_{\text{cosm}}$ is determined by two independently calibrated quantities: β_0 from SPARC rotation curves and

S_{cosm} from background observations. This cross-sector link is directly testable with Euclid and Rubin LSST $w(z)$ measurements. The linear form $w(a) = -\beta(S) \cdot a$ is the leading-order prediction; higher-order corrections from the full $S(a)$ -dynamics or the sigmoid gate function can produce $w_a \neq 0$ consistent with DESI DR2 data.

Regime Susceptibility Function

Definition and motivation

The coupling amplitude β measured at galactic scales (SPARC, $S_{\text{gal}} \approx 0.1$) and the cosmological amplitude required for $w_0 \approx -1$ differ. This is not a free parameter problem — it is a regime problem. The same physical coupling has different effective strength in different parts of the entropic continuum.

We define the *regime susceptibility function*:

$$T(S) \equiv \frac{S_0(1 - S_0)}{S(1 - S)}, \quad (6)$$

where S_0 is the reference regime (SPARC calibration, $S_0 \approx 0.1$). The cross-scale mapping is then:

$$\beta(S) = \beta_0 \cdot T(S) = \beta_0 \cdot \frac{S_0(1 - S_0)}{S(1 - S)}. \quad (7)$$

Properties of $T(S)$

Proposition 1 ($T(S)$ is self-regulating). *The product of the regime susceptibility and the effective energy density is constant:*

$$\beta(S) \cdot \rho_{\text{eff}}(S) = \beta_0 \cdot \frac{S_0(1 - S_0)}{S(1 - S)} \cdot S(1 - S) = \beta_0 S_0(1 - S_0) = \text{const.} \quad (8)$$

Proof. Direct substitution of equations (3) and (7). The $S(1 - S)$ terms cancel identically. $\square$

This self-regulation resolves the divergence problem: although $T(S) \rightarrow \infty$ as $S \rightarrow 0$ or $S \rightarrow 1$, the physically relevant quantity $\beta \cdot \rho_{\text{eff}}$ remains finite and constant throughout the entropic continuum. GR recovery at $S \rightarrow 0$ (Part 1, Theorem 2) is therefore not threatened by T .

Physical interpretation

$T(S)$ is the inverse of the system’s dynamical capacity. Near the thermodynamic boundaries, the system is “stiff” — small perturbations have large effects. At the midpoint $S = 0.5$, the system is maximally flexible — the same coupling produces the smallest observable response.

This mirrors the susceptibility of a physical system near a critical point, where the response function diverges as the order parameter approaches zero. In EFC, the “order parameter” is the entropic distance from either boundary, and $1/T(S) = S(1 - S)$ plays the role of the susceptibility denominator.

Table 1: Regime susceptibility at key epochs

Epoch	S	$S(1 - S)$	$T(S)/T(S_0)$	Physical domain
Singularity	$\rightarrow 0$	$\rightarrow 0$	$\rightarrow \infty$	Regulated by $\rho_{\text{eff}} \rightarrow 0$
CMB / L0	0.01	0.010	0.9	GR regime
Solar system / L1	0.05	0.048	1.9	PPN regime
Galactic (SPARC)	0.10	0.090	1.0	Reference (S_0)
Matter–DE transition	0.50	0.250	0.36	Peak capacity
Cosmic today	0.70	0.210	0.43	DE-dominated
Late de Sitter	0.90	0.090	1.0	Symmetric to SPARC
Altular limit	$\rightarrow 1$	$\rightarrow 0$	$\rightarrow \infty$	Regulated by $\rho_{\text{eff}} \rightarrow 0$

Cross-Scale Prediction for w_0

Inferring S_{cosm} from w_0

With $\beta_0 \approx 0.16$ (SPARC) and $S_0 \approx 0.1$, equation (5) at $a = 1$ gives:

$$w_0 = -\beta_{\text{cosm}} = -\beta_0 \frac{S_0(1 - S_0)}{S_{\text{cosm}}(1 - S_{\text{cosm}})}. \quad (9)$$

Setting $w_0 = -1$ (consistent with current observations) and solving for S_{cosm} :

$$S_{\text{cosm}}(1 - S_{\text{cosm}}) = \beta_0 S_0(1 - S_0) = 0.16 \times 0.09 = 0.0144. \quad (10)$$

The quadratic $S^2 - S + 0.0144 = 0$ has two solutions: $S \approx 0.985$ and $S \approx 0.015$. The low- S root ($S \approx 0.015$) corresponds to the early universe (GR regime); the late-universe solution is:

$$\boxed{S_{\text{cosm}}(z = 0) \approx 0.985.} \quad (11)$$

This places the present epoch very close to the Altular limit, consistent with the observed transition into a DE-dominated, asymptotically de Sitter phase. The growth-sector parameter n does not appear in this derivation.

Late-time observability and the coincidence question

The value $S_{\text{cosm}} \approx 0.985$ raises a natural question: why does the universe appear to be near a thermodynamic boundary precisely at the epoch of observation?

EFC provides a structural answer. The regime susceptibility $T(S)$ diverges near the boundaries, but the physical observable $\beta(S) \cdot \rho_{\text{eff}}(S) = \text{const}$ (Proposition 1). What *does* diverge near $S \rightarrow 1$ is the *ratio* of the dark energy signal to the matter signal:

$$\frac{\beta_{\text{cosm}}}{\beta_{\text{gal}}} = \frac{T(S_{\text{cosm}})}{T(S_0)} = \frac{S_0(1 - S_0)}{S_{\text{cosm}}(1 - S_{\text{cosm}})} \approx \frac{0.090}{0.0144} \approx 6.25. \quad (12)$$

This amplification means that entropic effects at cosmological scales become strongly *observable* precisely in the late-entropy regime. Observers in the early universe ($S \ll 0.5$) would measure $\beta_{\text{cosm}} \approx \beta_{\text{gal}}$ — no apparent DE. Observers in the regime $S \approx 0.985$ measure a large amplification — apparent DE dominance.

The coincidence is therefore not a fine-tuning: it is a *selection effect* arising from the structure of $T(S)$. We observe the universe in a DE-dominated phase because that is the phase in which the entropic signature of the Altular limit becomes detectable.

Remark 2 (Proximity to $S = 1$ does not imply thermodynamic completion). $S_{\text{cosm}} \approx 0.985$ does not mean the universe is “nearly finished.” The approach to $S = 1$ is asymptotic: $dS/dt \rightarrow 0$ as $\rho_{\text{eff}} = S(1 - S) \rightarrow 0$, giving an indefinitely long de Sitter-like phase with extremely slow evolution and no abrupt termination. The universe enters a regime of low dynamical capacity and long timescales, not heat death. Structural activity (star formation, galactic evolution) has already ceased in most of the volume; what remains is geometric expansion at effectively constant H .

S_{cosm} as a cross-sector prediction

Equation (11) is not a fit parameter. It is *inferred* from two independently calibrated quantities: $\beta_0 \approx 0.16$ from SPARC galactic rotation curves and $w_0 \approx -1$ from cosmological expansion measurements. The fact that these two entirely different observational streams yield a consistent S_{cosm} is a non-trivial cross-sector consistency check.

Euclid and Rubin LSST will measure w_0 and w_a at precision $\sigma(w_0) < 0.02$, $\sigma(w_a) < 0.1$. Combined with any future improvement in β_0 from SPARC-successor surveys, equation (9) becomes a precision test of the cross-scale mapping.

Regime mismatch as measurement theory

The cross-scale mapping implies a precise reinterpretation of cosmological measurements. An instrument calibrated at S_{inst} applied to a system at S_{sys} yields:

$$G_{\text{obs}} = G_{\text{eff}}(S_{\text{sys}}) \cdot \frac{S_{\text{sys}}(1 - S_{\text{sys}})}{S_{\text{inst}}(1 - S_{\text{inst}})}. \quad (13)$$

When $S_{\text{inst}} < S_{\text{sys}}$: $G_{\text{obs}} > G_{\text{eff}}$ — the observer infers more gravitational coupling than is physically present, and attributes the excess to dark matter.

When $S_{\text{inst}} > S_{\text{sys}}$: $G_{\text{obs}} < G_{\text{eff}}$ — expansion appears faster than local binding predicts, and the deficit is attributed to dark energy.

Both follow from equation (13). Neither requires new particles or fields. The apparent dark sector is the observational signature of regime-mismatch in a universe traversing the entropic continuum.

Discussion

The inversion of causal hierarchy

Standard cosmology (GR + Λ CDM) places mass-energy as primary: matter curves spacetime, spacetime drives expansion. EFC inverts this hierarchy:

1. Energy flow J^μ is primary.
2. Gravity is emergent: $G_{\text{eff}} = G_{\text{eff}}(S, \text{EF})$.
3. Expansion is not driven — it is the *relaxation* of binding as entropic capacity $S(1 - S)$ decreases toward the Altular limit.

Accelerated expansion is therefore not caused by a new form of energy pushing space apart. It is caused by the progressive weakening of the convergent flow that was holding structure together. The universe does not accelerate; it *decelerates less*.

Consistency with Parts 1–3

The results of this paper are consistent with the earlier series:

- **Recovery (Part 1):** At $S \rightarrow 0$, $T(S) \rightarrow \infty$ but $\beta \cdot \rho_{\text{eff}} = \text{const}$, so GR recovery is preserved.
- **Growth suppression (Part 2):** $\mu(a) = 1 - Bg(a; n)$ with $B > 0$ remains valid; the β -scaling does not alter the sign or form of μ .
- **Kill criteria (Part 3):** Kill Criterion 3 (S_8 convergence) and Kill Criterion 4 (gravitational slip) are unchanged.

Kill Criterion 5 (Dynamical dark energy). If Euclid measures $w(z) = -1$ to within precision $\sigma(w_0) < 0.02$ across the redshift range $z \in [0, 2]$, the dynamical prediction $w(a) = -\beta(S) \cdot a$ is falsified, and with it the regime susceptibility interpretation of dark energy. This criterion is independent of and complementary to Kill Criteria 1–4 of Part 3.

Observer selection and the coincidence problem. The apparent late-time proximity to $S \rightarrow 1$ is not a coincidence of cosmic timing. It is a consequence of observer selection: only in this regime does the entropic signal exceed measurement thresholds. In the early universe ($S \ll 0.5$), $T(S) \approx T(S_0)$ and $\beta_{\text{cosm}} \approx \beta_{\text{gal}}$ — no detectable DE signature. At $S \approx 0.985$, the amplification $\beta_{\text{cosm}}/\beta_{\text{gal}} \approx 6.25$ makes the entropic imprint of the Altular limit observationally dominant. We are not near the end of the universe; we are in the first epoch where the approach to that end is *measurable*.

What is not established

The following remain open:

1. **$T(S)$ from first principles.** Equation (6) is proposed on structural grounds. A derivation from the EFC action or Lagrangian is required.
2. **$S_{\text{cosm}}(z = 0)$ independently.** The present-epoch entropic state is constrained only indirectly via w_0 .

3. **$H(z)$ recovery.** Equation (5) gives $w(a)$ but not $H(z)$ directly. The full Friedmann-level derivation remains open (Part 1, Remark 1).
4. **Regulation at boundaries.** While Proposition 1 shows that $\beta \cdot \rho_{\text{eff}}$ is finite, the behaviour of individual terms near $S = 0$ and $S = 1$ should be examined in the full field-equation context.

Conclusions

We have proposed a regime susceptibility function $T(S) = S_0(1 - S_0)/[S(1 - S)]$ that emerges directly from the $S(1 - S)$ structure already present in EFC, introducing no new fields or free parameters. Its key properties are:

1. **Dimensionless and parameter-free:** derived from the same entropy field $S(a)$ used throughout Parts 1–3.
2. **Exactly self-regulating:** $\beta(S) \cdot \rho_{\text{eff}}(S) = \beta_0 S_0(1 - S_0) = \text{const}$ (Proposition 1), preserving GR recovery at all boundaries.
3. **Cross-scale consistent:** bridges SPARC-calibrated $\beta_0 \approx 0.16$ to cosmological $w_0 \approx -1$ with a measurable amplification factor ≈ 6.25 .
4. **Unifying:** dark matter and dark energy are the convergent and divergent phases of a single entropic continuum; their apparent magnitudes are regime-mismatch artefacts, not independent substances.
5. **Falsifiable:** Kill Criterion 5 — if Euclid measures $w(z) = -1$ at $\sigma(w_0) < 0.02$, the mechanism is falsified.

The full chain is now:

$$S(a) \longrightarrow T(S) \longrightarrow \beta(S) \longrightarrow w(a) \longrightarrow H(z)\text{-dynamics (candidate)}$$

and connects to the same growth-sector seal, SPARC calibration, and validation ledger established in Parts 1–3.

EFC is not yet a complete cosmological theory. But the structural gap between galactic and cosmological scales has a candidate closure mechanism that is self-consistent, parameter-free, and testable. That is what the next stage of data requires.

References

- [1] Magnusson, M. (2026). EFC Part 1: Recovery Conditions and the Λ CDM Limit. doi:10.6084/m9.figshare.31970886
- [2] Magnusson, M. (2026). EFC Part 2: Field Equations and Observable Mapping. doi:10.6084/m9.figshare.31970898
- [3] Magnusson, M. (2026). EFC Part 3: Validation Ledger and Falsification Protocol. doi:10.6084/m9.figshare.31970904
- [4] Magnusson, M. (2026). Falsifiability Protocol for EFC. doi:10.6084/m9.figshare.31251277
- [5] Magnusson, M. (2026). Minimal EFC Effective Field Theory Ansatz. doi:10.6084/m9.figshare.31562200

- [6] Magnusson, M. (2026). Covariant EFT for Entropy-Driven Gravitational Modification. [doi:10.6084/m9.figshare.31878334](https://doi.org/10.6084/m9.figshare.31878334)
- [7] Magnusson, M. (2026). Regime-Dependent Validity in EFC: Evidence from SPARC. [doi:10.6084/m9.figshare.31007248](https://doi.org/10.6084/m9.figshare.31007248)
- [8] Magnusson, M. (2026). $R(k, S)$ as a Regime Response Surface in Energy-Flow Cosmology. [doi:10.6084/m9.figshare.31211437](https://doi.org/10.6084/m9.figshare.31211437)
- [9] Magnusson, M. (2026). EFC Ontological Foundations v1.0.2. [doi:10.6084/m9.figshare.31223668](https://doi.org/10.6084/m9.figshare.31223668)